

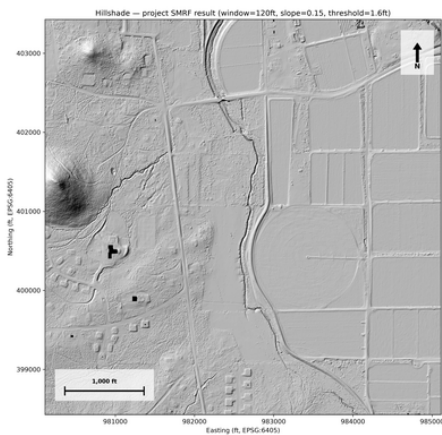
# LiDAR Bare-Earth DEM — QC Summary

San Xavier District, Pima County, AZ · USGS 3DEP, QL2, Feb 2015 · 574 acres · EPSG:6405 (International Feet)

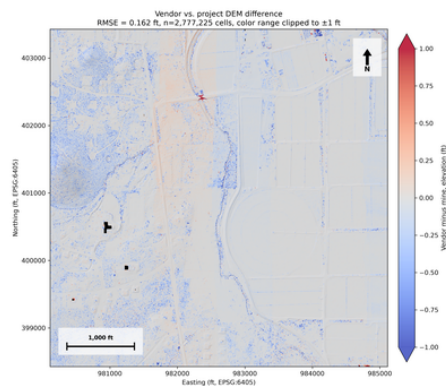
Bare-earth DEM built from raw LiDAR returns — not the vendor's delivered classification — and independently QC'd against the vendor product, flight-line geometry, and the source collection's own signed accuracy assessment. **Method:** ELM + statistical outlier removal + SMRF ground classification (window 120 ft, slope 0.15, threshold 1.6 ft) → 3 ft IDW bare-earth surface.

## Key Findings

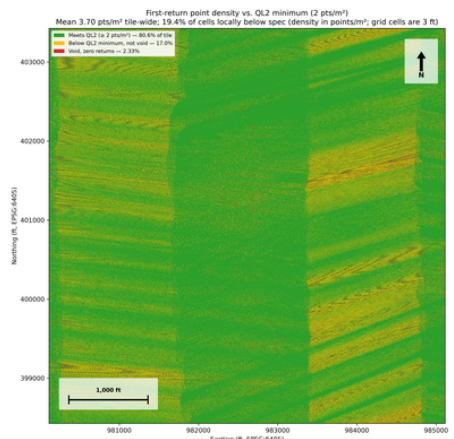
- **Accuracy — 0.162 ft RMSE** vs. the vendor's own delivered ground surface, clearing the ASPRS QL2 non-vegetated threshold ( $\leq 0.328$  ft) with margin.
- **Flight-line offset — ~0.12 ft systematic** vertical offset between two overlapping flight lines in the source collection: a real calibration artifact in the data, not a processing error.
- **Coverage — 19.4% of cells locally below QL2 minimum** despite a passing tile-wide average (3.70 pts/m<sup>2</sup>) — local density needs its own check.
- **Roof vs. pad — resolved.** Crisp rectangular features in the residential block are graded concrete pads (+2.85 ft), not unremoved roof returns (would show 8–15 ft); corroborated against current orthoimagery.



The deliverable: bare-earth hillshade, SMRF-classified from raw returns



Accuracy: 0.162 ft RMSE vs. vendor's own classification (clears QL2)



Coverage: meets QL2 on average, 19.4% of cells locally short

Full 18-page report: [output/reports/qc\\_report.pdf](#) · Source memo: [output/reports/qc\\_memo.md](#) · Data: [USGS\\_LPC\\_Eastern\\_Pima\\_County\\_Lidar\\_980398.laz](#)